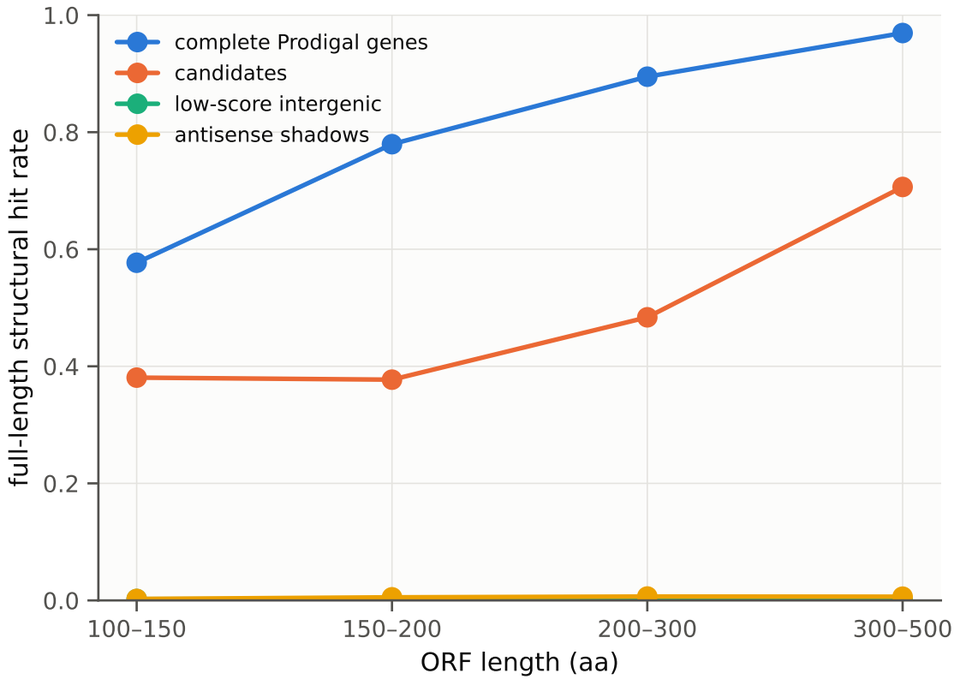
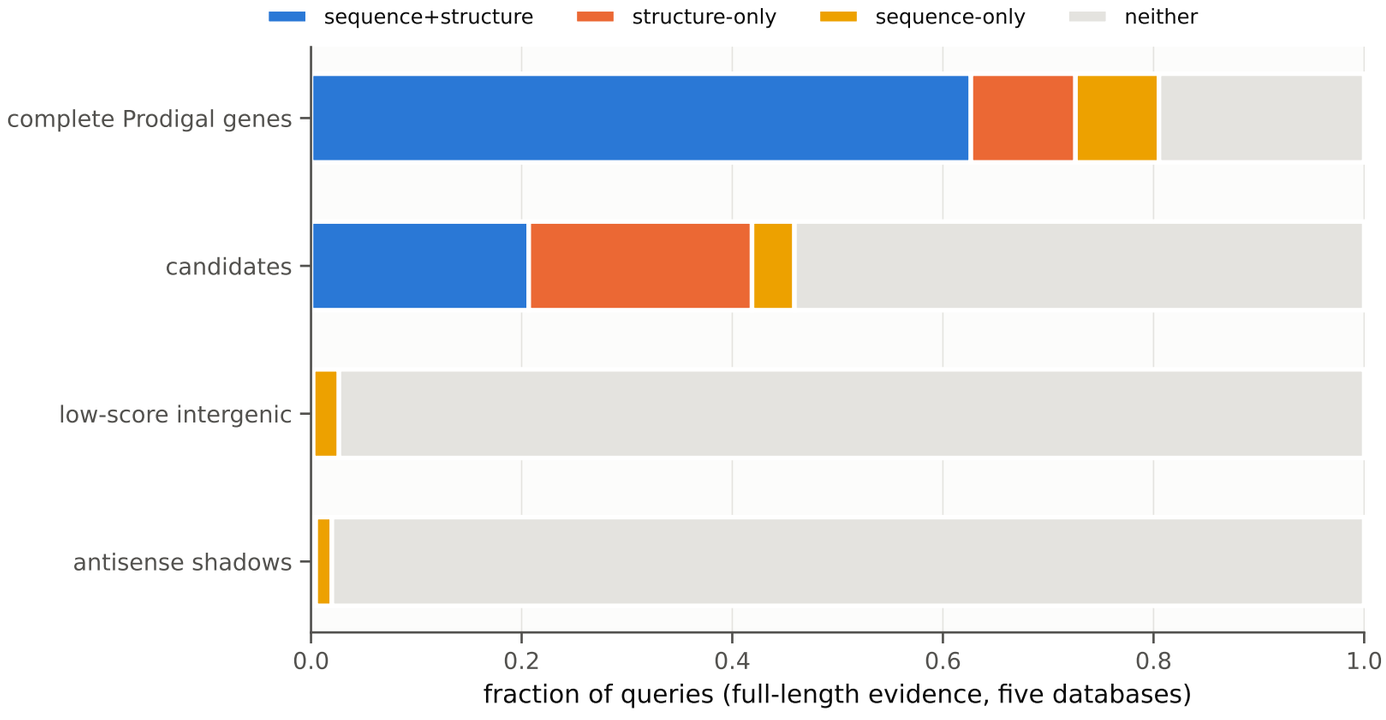
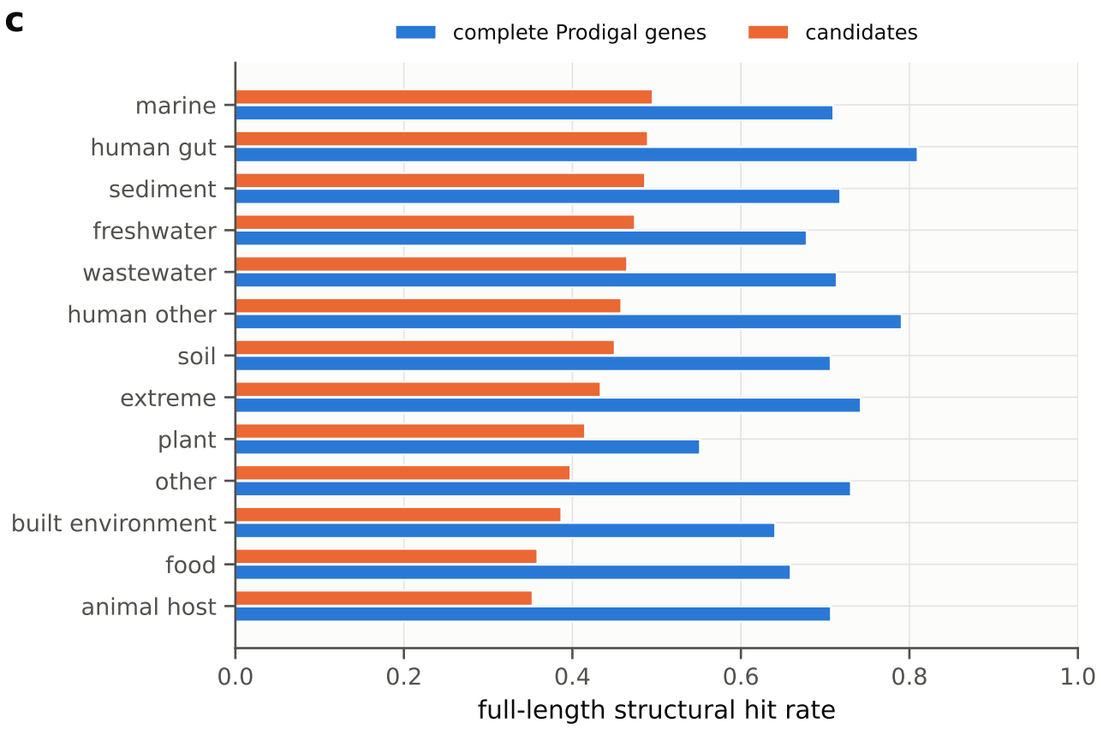
a


complete Prodigal genes
candidates
low-score intergenic
antisense shadows
full-length structural hit rate
1.0
0.8
0.6
0.4
0.2
0.0
100–150
150–200
200–300
300–500
ORF length (aa)

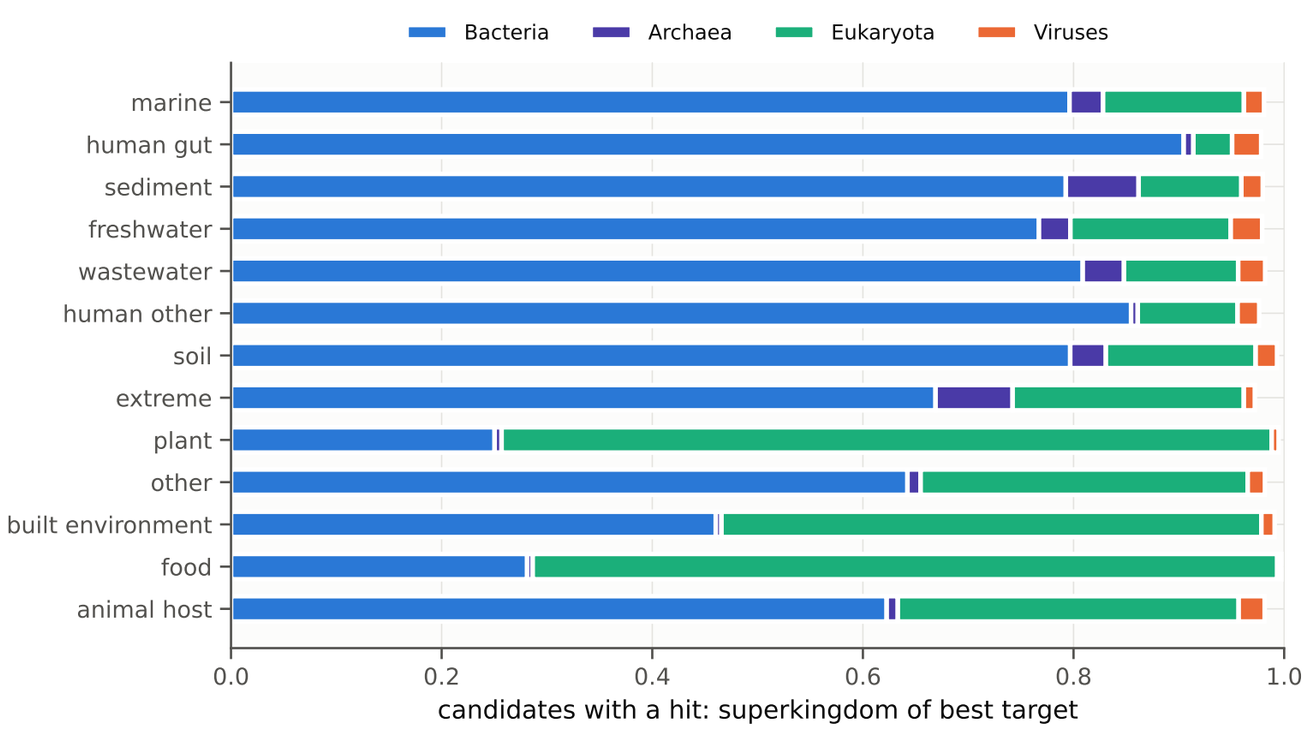
b


sequence+structure
structure-only
sequence-only
neither
complete Prodigal genes
candidates
low-score intergenic
antisense shadows
0.0
0.2
0.4
0.6
0.8
1.0
fraction of queries (full-full-length evidence, five databases)

c


complete Prodigal genes
candidates
marine
human gut
sediment
freshwater
wastewater
human other
soil
extreme
plant
other
built environment
food
animal host
0.0
0.2
0.4
0.6
0.8
1.0
full-length structural hit rate

d


Bacteria
Archaea
Eukaryota
Viruses
marine
human gut
sediment
freshwater
wastewater
human other
soil
extreme
plant
other
built environment
food
animal host
0.0
0.2
0.4
0.6
0.8
1.0
candidates with a hit: superkingdom of best target